

TOPOGRAPHY

The hip joint is a synovial joint of ball-and-socket (multiaxial spheroidal) type between the head of the femur and the cup-shaped (cotyloid) acetabulum; its centre lies a little inferior to the middle one-third of the inguinal ligament.

Articular surfaces

The femoral head can be landmarked from the surface roughly 2–4 cm superior to an approximate midpoint of a line joining the superior margin of the greater trochanter to the pubic tubercle. The articular surfaces of the femoral head and the acetabulum are reciprocally curved but neither coextensive nor completely congruent (Fig. 81.1; see Fig. 80.16). The close-packed position of the hip joint is one of full extension, with slight abduction and medial rotation (see also Table 5.4). This position winds up most of the joint's capsular ligaments, which can provide an important element of articular stability. The femoral head is covered by articular cartilage, except over the rough pit where the ligament of the head of the femur is attached. Anteriorly, the cartilage extends laterally over a small area on the adjoining neck. Articular cartilage is, generally, thicker centrally than peripherally. Cartilage thickness is maximal anterosuperiorly in the acetabulum and anterolaterally on the femoral head, the two areas that correspond to the principal load-bearing areas within the joint. The acetabular articular surface, the lunate surface, is an incomplete ring, broadest anterosuperiorly where the pressure of body weight falls in the erect posture, and narrowest in its pubic region. It is deficient inferiorly opposite the acetabular notch. The lunate surface is covered by articular cartilage, which is thickest where the surface is broadest. The acetabular fossa, the central non-articular area in the floor of the acetabulum, is devoid of cartilage but contains fibroelastic fat largely covered by synovial membrane. The acetabular labrum, a fibrocartilaginous rim attached to the acetabular margin, serves to deepen the acetabulum and bridges the acetabular notch by attaching to the peripheral edge of the transverse acetabular

ligament. The acetabular labrum is triangular in section, attached by its base to the acetabular margin and blending with adjacent articular cartilage; its acute free edge projects beyond the acetabular margin. The diameter of the acetabular cavity is constricted by the labral rim, which embraces the femoral head, thereby contributing to the stability of the articulation. The acetabular labrum increases the surface area for joint contact as well as sealing the joint, allowing optimal intra-articular distribution of synovial fluid (Ferguson et al 2003, Cadet et al 2012). It thus facilitates nutrition of the articular cartilage and helps to reduce intra-articular friction (Safran et al 2011, Song et al 2012). Nerve endings found within the labrum suggest this structure may be a source of proprioception or, when injured, of pain (Kim and Azuma 1995). A torn acetabular labrum can be detected using MRI, to identify the site and severity of the lesion (Fig. 81.2).



Acetabular labrum and impingement injury



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Fibrous capsule

The fibrous capsule of the hip joint is strong and dense (Fig. 81.4). It is attached superiorly to the acetabular margin 5–6 mm medial to the labral attachment, anteriorly to the outer labral aspect and, near the acetabular notch, to the transverse acetabular ligament and the adjacent rim of the obturator foramen. From its acetabular attachment, it extends laterally to surround the femoral head and neck, and is attached anteriorly to the intertrochanteric line, superiorly to the base of the femoral neck, posteriorly 1 cm superomedial to the intertrochanteric crest, and inferiorly to the femoral neck near the lesser trochanter. Anteriorly, many fibres ascend along the neck as longitudinal retinacula, containing blood vessels for both the femoral head and the neck. The capsule is thicker anterosuperiorly, where maximal stress occurs, particularly while standing with the hip extended. Posteroinferiorly, it is relatively thin and loosely attached. The capsule as a whole has two sets of fibres,



Fig. 81.1A Anteroposterior radiograph of an adult female pelvis. 1, sacral promontory; 2, margin of anterior sacral foramen; 3, linea terminalis; 4, fovea for ligament of head; 5, lesser trochanter of left femur; 6, left ischial tuberosity; 7, left obturator foramen; 8, coccyx; 9, left anterior inferior iliac spine; 10, gas in sigmoid colon. (B, continued online)



Fig. 81.1B Anteroposterior radiograph of an adult male pelvis (the hip joint is normal). 1, ossification in right iliofemoral ligament; 2, pubic symphysis; 3, inferior margin of left sacroiliac joint; 4, left anterior superior iliac spine.

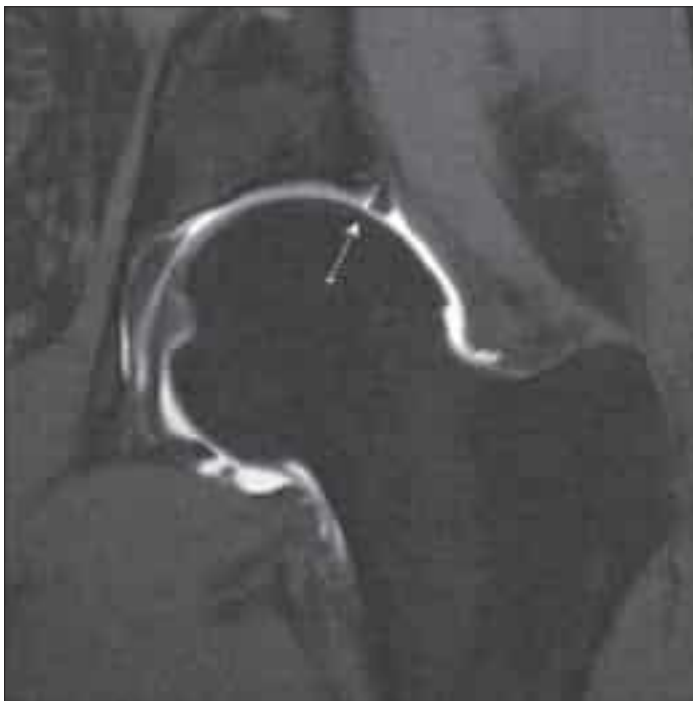


Fig. 81.2 T1-weighted fat-saturated MR arthrogram of the left hip joint (coronal section). The arrow points to a torn acetabular labrum at the anterosuperior margin of the acetabulum.



Fig. 81.3 Pre-surgical radiograph of a hip with a cam deformity of the femoral head-neck junction (arrow). Note the loss of sphericity in the femoral head and the loss of the tapering of the neck. (Image courtesy of John T Heinrich MD, Milwaukee Orthopaedic Group Limited, Milwaukee, WI, USA.)

The shape of the bones of the hip normally limits large and potentially damaging contact stress on adjacent connective tissues, especially during the extremes of movement. Consider, for example, the normal shape of the femoral head and neck region. Similar to the shape of a light bulb, the spherical femoral head narrows gradually at its junction with the neck of the femur. This tapering limits large contact stress between the femoral head and the margin of the acetabulum. However, tapering may be lost as a result of natural variation or underlying pathology, resulting in a specific deformity that has been referred to as a cam deformity because the head and neck of the femur, together, resemble a camshaft. Although often subtle, this deformity may be detected radiographically (see arrow in [Fig. 81.3](#)). In some patients with a cam deformity, extreme and repeated motions of the hip, typically involving combinations of flexion, medial rotation and adduction, can

create damaging impingement between the hard and bulbous femoral neck and the softer acetabular labrum. The anterosuperior section of the labrum is particularly vulnerable to this type of injury, which is a painful and often motion-limiting condition referred to as femoral-acetabular impingement syndrome. The repeatedly traumatized section of the labrum can become enlarged, fragmented or torn. When the acetabular labrum is torn, a hip joint may become unstable as a consequence of the loss of the natural suction seal around the joint. In some advanced cases, the impingement trauma may create lesions at the chondrolabral junction, resulting in damage to the adjacent articular cartilage, potentially predisposing the hip joint to degenerative arthritis (osteoarthritis). Treatment may involve arthroscopic repair of the acetabular labrum and cartilage, as well as reshaping the area of the femoral head and neck responsible for the impingement.

circular and longitudinal. The circular fibres (zona orbicularis) are internal and form a collar round the femoral neck; although partly blended with the pubofemoral and ischiofemoral ligaments, these fibres are not directly attached to bone (see Fig. 81.4). Externally, longitudinal fibres are most numerous in the anterosuperior region, where

they are reinforced by the iliofemoral ligament. The capsule is also strengthened inferiorly by the pubofemoral ligament, and posteriorly by the ischiofemoral ligament. Externally the capsule is rough, covered by muscles and tendons, and separated anteriorly from psoas major and iliacus by a bursa. The capsular attachment to the femur lies well distal to the growth plate of the femoral head both anteriorly and posteriorly, and so the upper femoral epiphysis is entirely intracapsular. The capsular attachment intersects the growth plate of the greater trochanter on the superior surface of the base of the neck.

Iliocapsularis



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Relations The joint capsule is surrounded by muscles (Fig. 81.5). Anteriorly, lateral fibres of pectineus separate the capsule from the femoral vein. Lateral to this, the tendon of psoas major, with iliacus lateral to it, descends en route to the lesser trochanter, partly separated from the capsule by a bursa. The femoral artery is anterior to the tendon of psoas major, and the femoral nerve lies deep in a groove between the tendon and iliacus. More laterally, the straight head of rectus femoris crosses the joint with a deep layer of the iliotibial tract, which blends with the capsule under the lateral border of the muscle. Superiorly, the reflected head of rectus femoris contacts the capsule medially, while gluteus minimus covers it laterally, being closely adherent. Inferiorly, medial fibres of pectineus adjoin the capsule and, more posteriorly, obturator externus spirals obliquely to its posterior aspect. Posteroinferiorly, the capsule is covered by the tendon of obturator externus, separating it from quadratus femoris and accompanied by an ascending branch of the medial circumflex femoral artery. Superior to this, the tendon of obturator internus and the gemelli contact the joint capsule, separating it from the sciatic nerve. The nerve to quadratus femoris is deep to the obturator internus tendon and descends medially on the capsule. Superior to this, the posterior surface of the joint is crossed by piriformis.

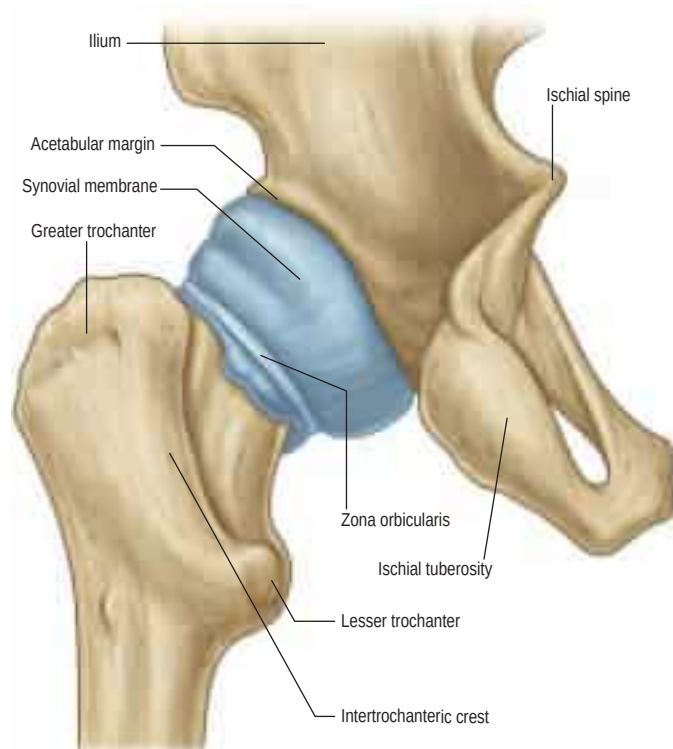


Fig. 81.4 Synovial cavity of the left hip joint (distended): posterior aspect.

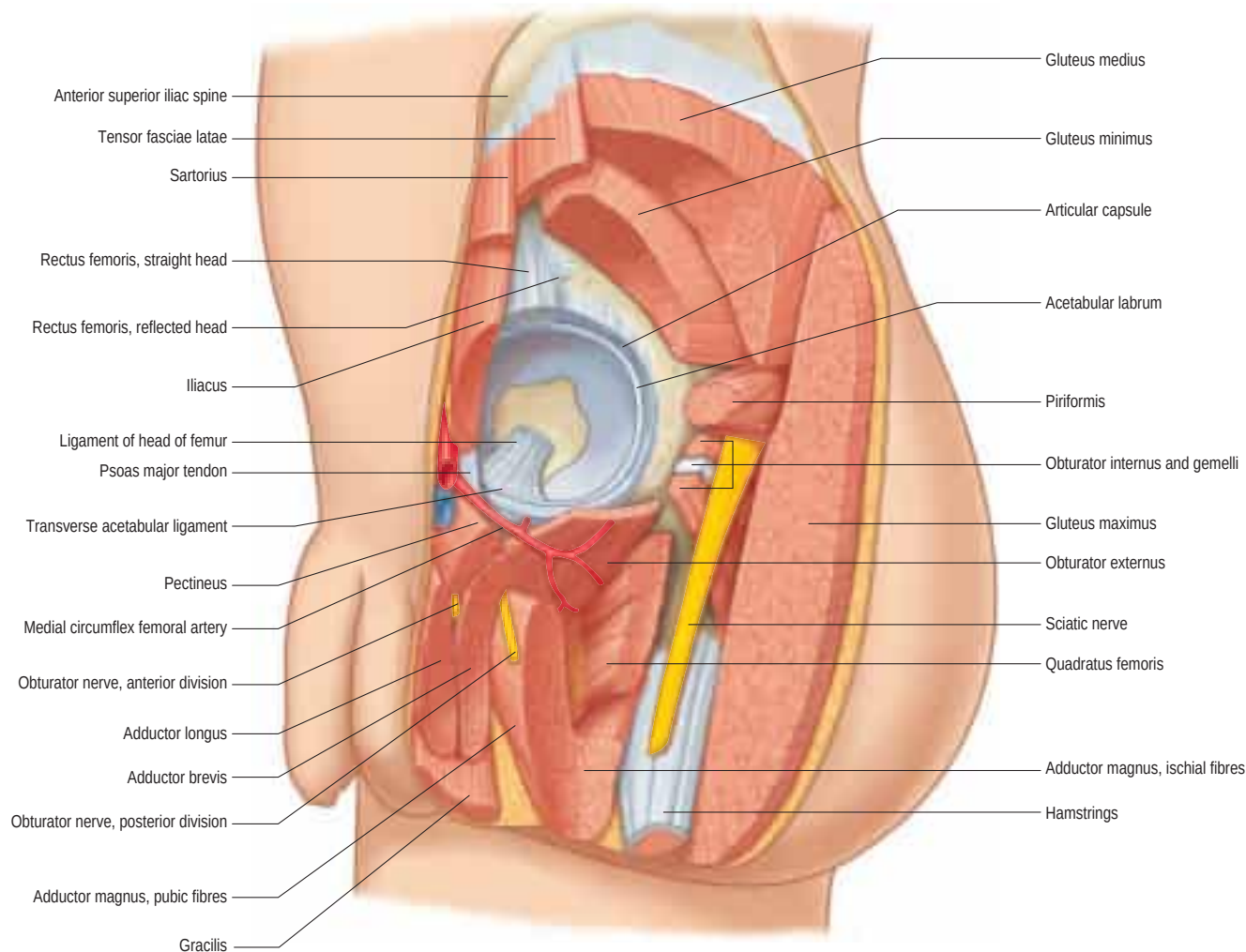


Fig. 81.5 Schematic dissection to display the structures surrounding the hip joint. The femoral head has been disarticulated and removed.

Iliocapsularis (also called iliacus minor or iliotrochantericus) is a little-known but constant muscle that lies immediately deep to the tendon of iliopsoas and overlies the anterior hip capsule ([Babst et al 2011](#), [Ward et al 2000](#)). Typically present as just a few strands of fibres, it originates from the anteromedial hip capsule and anteroinferior iliac spine, and is attached just distal to the lesser trochanter. Although the function of the muscle is uncertain, it may draw the capsule taut as a way of reducing its impingement during hip flexion.

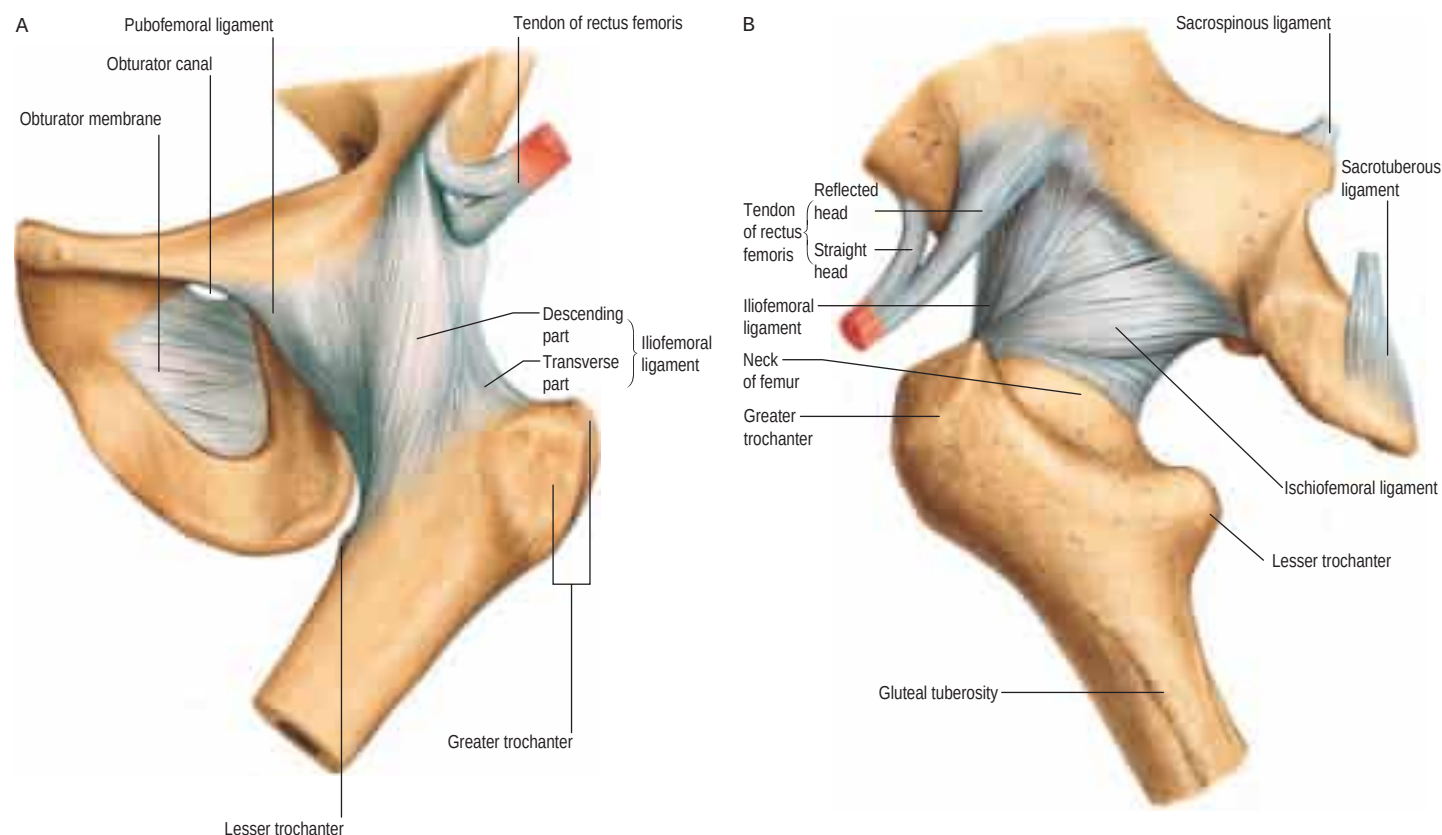


Fig. 81.6 Ligaments of the hip joint. **A**, Anterior aspect; **B**, Posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Ligaments

The ligaments of the hip joint are the iliofemoral, pubofemoral, ischiofemoral and transverse acetabular ligaments, and the ligament of the head of the femur (Fig. 81.6). As the hip moves, the capsular ligaments wind and unwind, tightening around the hip, and affecting stability, excursion and joint capacity (Fuss and Bacher 1991). Joint capacity is maximal when the hip joint is held in a partially flexed and abducted position; a patient with an effusion in the hip joint is therefore most comfortable when the joint is held in this position.

Iliofofemoral ligament The iliofofemoral ligament is very strong and is shaped like an inverted Y, lying anteriorly and intimately blended with the capsule. Its apex is attached between the anterior inferior iliac spine and acetabular margin, its base to the intertrochanteric line. It is composed of a medially placed, thicker descending part and a laterally placed, transverse part, which is typically distinguishable anteriorly. The obliquely disposed transverse part is attached to a tubercle at the superolateral end of the intertrochanteric line, while the vertically oriented descending part reaches the inferomedial end of the line.

Pubofemoral ligament The pubofemoral ligament is triangular, with a base that is attached to the iliopubic ramus, superior pubic ramus, obturator crest and membrane. It blends distally with the capsule and deep surface of the descending part of the iliofemoral ligament and has been described as consisting of multiple crura (Fuss and Bacher 1991).

Ischiofemoral ligament The ischiofemoral ligament thickens the posterior aspect of the capsule. The central part spirals superolaterally from the ischium, where it is attached posteroinferior to the acetabulum, then courses posterior to the femoral neck to attach distally to the greater trochanter deep to the iliofemoral ligament. Some fibres blend with the zona orbicularis. Some of the more inferior fibres of the ischiofemoral ligament embrace the posterior circumference of the femoral neck.

Transverse acetabular ligament The transverse acetabular ligament (see Fig. 80.4) is continuous peripherally with the labrum but does not possess any cartilage cells. Its strong, flat fibres cross the acetabular notch, forming a foramen through which vessels and nerves pass to enter the joint.

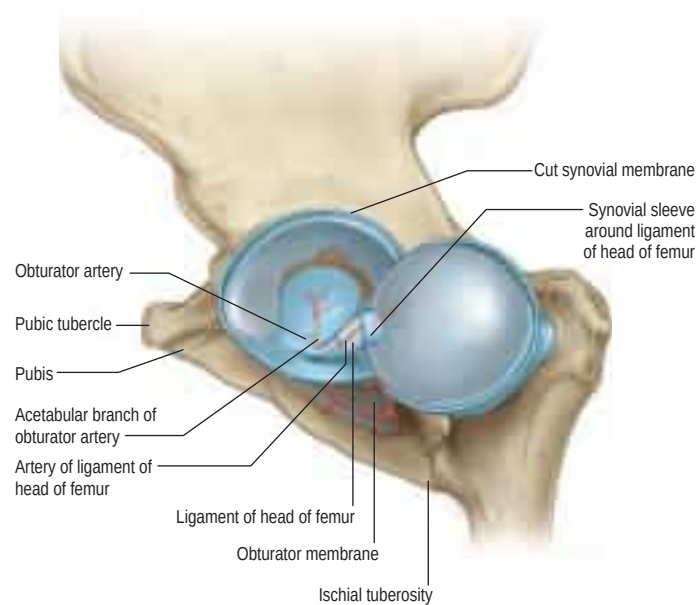


Fig. 81.7 Interior aspect of the hip joint to show the ligament of the head of the femur. (With permission from Drake RL, Vogl AW, Mitchell A, et al (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

Ligament of the head of the femur The ligament of the head of the femur is a triangular and somewhat flattened band of connective tissue (see Fig. 81.5, Fig. 81.7). Its apex is attached anterosuperiorly in the fovea for the ligament of the head of the femur while its base is attached principally to both edges of the acetabular notch, between which it blends with the transverse acetabular ligament. It also receives weaker contributions from the margins of the acetabular fossa, is ensheathed by synovial membrane and varies in strength. Occasionally, the synovial sheath exists by itself in the absence of the ligament; rarely, both the ligament and sheath are absent.

Although the ligament of the head of the femur is known primarily for lending structural support to the blood vessels that travel to and

from the femoral head, speculation exists that it may also have a role in stabilizing the hip joint *in utero*. In the adult, it is mostly elongated and assumed to be taut when the hip joint is semi-flexed, laterally rotated, and adducted, the position where the capsular ligaments, as a whole, offer least stability to the joint (Martin et al 2012). The degree to which the ligament of the head of the femur offers functional stability to the adult hip is uncertain, although it is likely to be secondary to that provided by the capsular ligaments and muscle activation.

Medical interest in the ligament of the head of the femur has increased with the advancing technology associated with medical imaging and arthroscopic hip surgery (Botser et al 2011, Gray and Villar 1997; Fig. 81.8). Free nerve endings have been identified within this ligament, suggesting that this structure may be a source of pain when it is injured or degenerating (Leunig et al 2000). The afferent innervation in the healthy state may serve as a source of proprioception or protection of the joint (Bardakos and Villar 2009).

Synovial membrane

Starting from the femoral articular margin, the synovial membrane covers the intracapsular part of the femoral neck, and then passes to the internal surface of the capsule to cover the acetabular labrum, the ligament of the head of femur and fat in the acetabular fossa. It is thin on the deep surface of the iliofemoral ligament where it is compressed against the femoral head, and sometimes is even absent here.

Bursae

The hip joint may communicate with the subtendinous iliopectineal bursa through a circular aperture between the pubofemoral ligament and the descending part of the iliofemoral ligament. More distant bursae are associated with the tendons of distal attachment of glutei medius and minimus at the greater trochanter, and between gluteus maximus and vastus lateralis.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Articular arteries are branches from the obturator, medial circumflex femoral, and superior and inferior gluteal arteries; they form the cruciate and trochanteric anastomoses (see Fig. 80.18). There is a corresponding venous drainage.

Lymphatics from the anterior aspect of the hip joint drain to the deep inguinal nodes, while those from the medial and posterior aspects of the joint travel with the obturator and gluteal arteries, respectively, to reach the internal iliac nodes.

INNERVATION

The anterior capsule of the hip joint is innervated by the femoral, obturator and accessory obturator (when present) nerves. The posterior capsule is innervated by the sciatic and superior gluteal nerves, and by the nerve to quadratus femoris (Birnbbaum et al 1997, Gardner 1948).

Hip and groin pain

Pain emanating from the hip and groin has many possible causes. True pain from the hip joint is typically felt deep in the crease of the groin. Pain over the lateral aspect of the hip usually suggests a local problem such as trochanteric bursitis. Pain in the buttock is only occasionally derived from the hip joint; most frequently, it is referred from the lumbosacral spine or may be caused by vascular claudication. Pain in the groin may be associated with inguinal or femoral hernias. In the younger age group, it afflicts those engaged in sports, especially running ball sports; the pain is sometimes due to femoro-acetabular impingement or to hernia but most often is related to adductor longus tendinitis ('groin strain'). Attritional chronic overload of the anterior abdominal wall at its attachments to the pubis and inguinal ligament, or of external oblique at the superficial inguinal ring, is a common occurrence. Rarely, problems may occur at the pubic symphysis.

FACTORS MAINTAINING STABILITY

The hip joint is normally very stable. The femoral head is closely fitted to the acetabulum in an area exceeding half a sphere, and is embraced

closely by the acetabular labrum, which restrains it in the socket. In addition, a 'vacuum effect' is present. The thick capsule is reinforced by the three major ligaments: iliofemoral, pubofemoral and ischiofemoral. The iliofemoral ligament is the strongest, and is progressively tightened when the femur extends to the line of the trunk. The pubofemoral and ischiofemoral ligaments also tighten when this happens and, as the joint approaches the close-packed position, resistance to an extending torque increases rapidly. The transverse acetabular ligament and ligament of the head of the femur also contribute to stability. Only slight separation of the articular surfaces can be achieved by strong traction on the joint. To aid insertion of an arthroscope, a needle is first inserted into the joint to eliminate the suction effect; the joint will subsequently open sufficiently with traction. Traumatic dislocation usually occurs only when the joint is subjected to extreme force.

BIOMECHANICS OF THE HIP JOINT

The transition from quadrupedal to bipedal gait was a considerable biomechanical milestone in the evolution of *Homo sapiens*. The biomechanical features of the human lower limb, whose function is primarily to allow stance and bipedal propulsion, are very different from those of the quadrupedal lower limb, and these differences are reflected in the anatomy and biomechanics of the human hip, knee, ankle and foot joints.

The acetabulum and femoral head form a multiaxial spheroidal ('ball and socket') joint, which allows relatively unhindered motion in three degrees of freedom, i.e. flexion/extension, abduction/adduction and medial/lateral rotation. This articulation is innately limited in its capacity for translational motion in anteroposterior, transverse and vertical directions.

Femur

The femur is essentially a tubular structure with distortions that consist of bows and twists. The most notable is the anterior bow in its mid portion, where the radius of curvature is relatively constant along the length of the femoral shaft. In the coronal (frontal) plane, the femoral neck is inclined obliquely to the shaft at an angle of about 135° (range 120–140°; Fig. 81.10). Although the neck-shaft angle (collo-diaphyseal angle; Mikulicz angle) and neck length are variable, the centre of the neck in the coronal plane is at the level of the apex of the greater trochanter. In the axial (transverse) plane, the femoral neck is anteverted, i.e. rotated anteriorly relative to the posterior surfaces of the femoral condyles; in the adult, this angle is 10–15° (Fig. 81.11A). Excessive anteversion may exist when this angle is significantly greater than 10–15° (Fig. 81.11B). At birth, the angle of anteversion is typically about 35–40°. As the child develops, forces from muscles and gravity cause the angle of anteversion to decrease gradually, approaching 15° by young adulthood. Excessive anteversion usually persists beyond

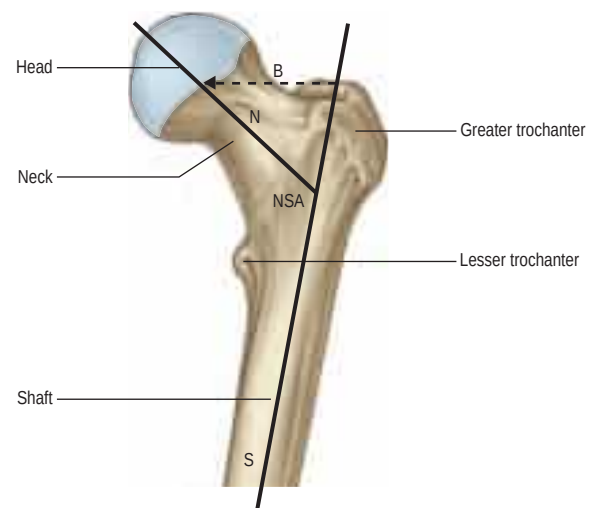


Fig. 81.10 The neck-shaft angle (NSA) between the long axis of the femoral shaft (S) and the axis of the femoral neck (N) is on average 135° (range 125–140°). In addition, in most hips, a line nearly perpendicular to S from the tip of the greater trochanter (B) passes through the centre of the femoral head. This approximation can be utilized in judging the position of the femoral osteotomy in hip arthroplasty.

The blood supply to the hip in the child is different from that in the adult and this has very important implications clinically. Since the growth plate is avascular, the only blood supply to the femoral head is conveyed through the retinacular vessels, which are mainly tributaries of the medial circumflex femoral artery. These retinacular arteries enter the joint capsule in the trochanteric fossa and ascend the femoral neck on its outer aspect superiorly and posteriorly. They pass around the edge of the growth plate (physis) and enter the femoral head (Fig. 81.9). A minor contribution to the vascular supply comes from the ligament of

the head of the femur. Any damage to these vessels (e.g. trauma, infection) can result in the devastating complication of avascular necrosis of the femoral head. The growth plate represents a line of weakness and predisposes to fracture from injury. Acute injuries affecting the capital epiphysis are uncommon. A more chronic fracture through the capital epiphysis resulting in slipped capital femoral epiphysis, where the femoral head epiphysis displaces posteriorly off the femoral neck, occurs in pubescent adolescents, especially males (see Fig. 81.9 and p. 1353).

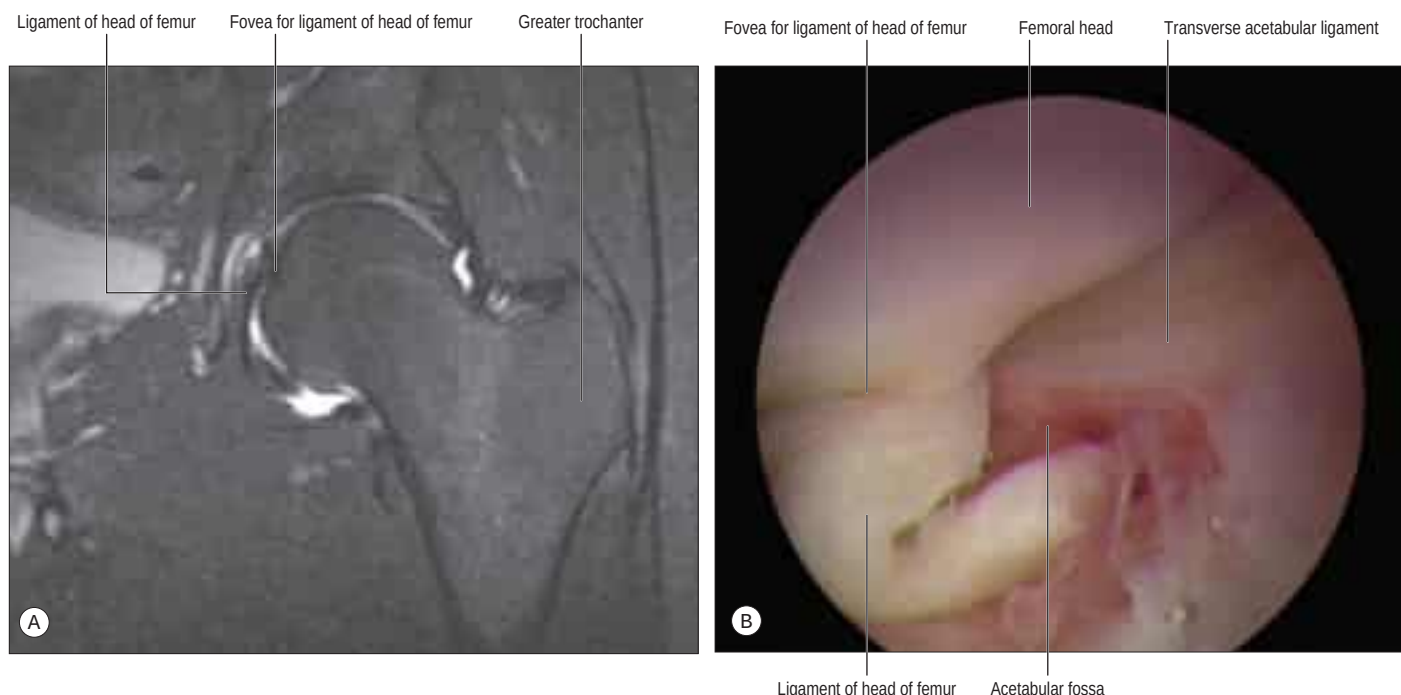


Fig. 81.8 An intact ligament of the head of the femur in a left adult hip. **A**, T2-weighted fat-saturated MR arthrogram of the hip joint (coronal section). (Courtesy of Michael O'Brien MD, Wisconsin Radiology Specialists, Milwaukee, WI, USA.) **B**, Photograph taken through an arthroscope of a slightly distracted hip joint. (Courtesy of John T Heinrich MD, Milwaukee Orthopaedic Group Limited, Milwaukee, WI, USA.)

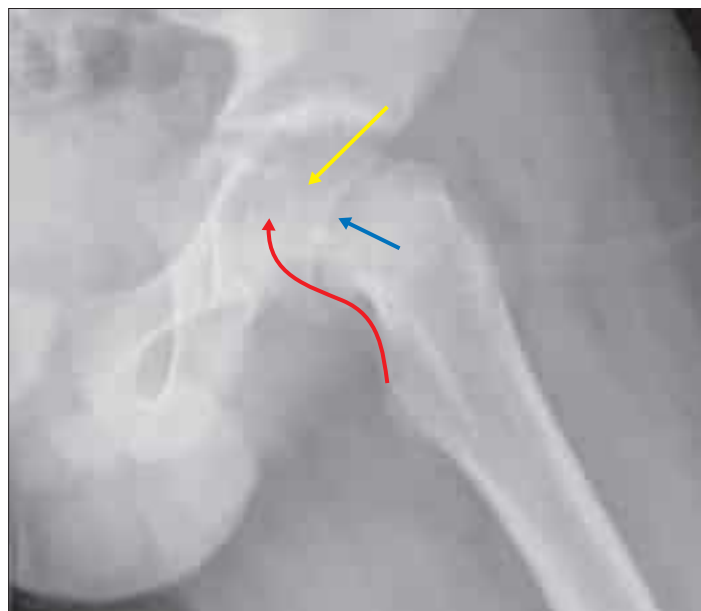


Fig. 81.9 Radiograph of the left hip of a 14-year-old boy. The yellow arrow shows posterior displacement of the femoral head in the slipped capital femoral epiphysis. The red arrow indicates the course of the retinacular vessels along the posterior aspect of the femoral neck. The blue arrow indicates the physis (growth plate). (Courtesy of Mr Christopher Edward Bache.)

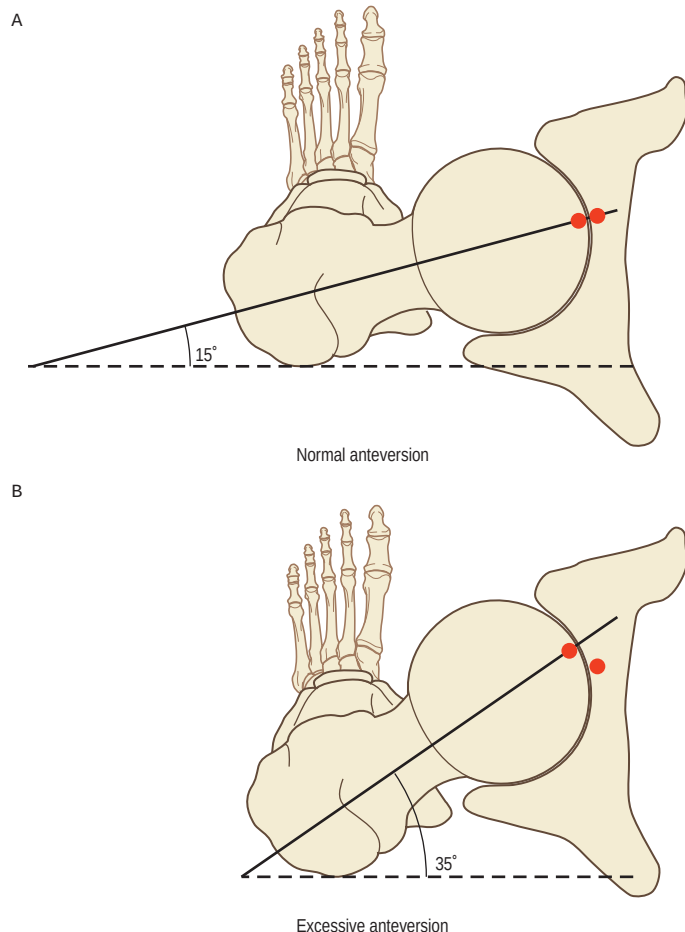


Fig. 81.11 **A**, Superior view of a proximal femur with normal anteversion. **B**, Superior view of a proximal femur with excessive anteversion. (With permission from Neumann D, *Kinesiology of the Musculoskeletal System*, 2nd ed, Elsevier, Mosby. Copyright 2010.)

childhood in cases of abnormal motor development, such as can be seen with cerebral palsy.

The degree of anteversion affects many aspects of lower limb biomechanics, including the moment (lever) arms of the hip abductor and medial rotator muscles (the perpendicular distance from the centre of rotation of the femur to the line of action of the resultant muscle force); patellar tracking (the motion of the patella relative to the femur); and foot orientation.

Acetabulum

The acetabulum consists of the confluence of the ilium, ischium and pubis at the triradiate cartilage. By itself, the acetabulum covers an area slightly less than a hemisphere; it is deepened by the acetabular labrum. The degree of acetabular anteversion is about 17° (16° males and 18° females) (Reikeras et al 1983). In the coronal plane, the acetabular axis is inclined inferiorly approximately 45° from the horizontal.

MOVEMENTS



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Muscles producing hip joint movements

Flexion is primarily produced by psoas major, iliacus and rectus femoris, assisted by pectineus, tensor fasciae latae and sartorius. The adductors, particularly adductor longus and brevis, also assist, especially when the hip is near full extension. Effective action of the hip flexor muscles requires strong synergistic activation of the abdominal muscles to stabilize the pelvis.

Extension is produced by gluteus maximus, biceps femoris, semitendinosus, semimembranosus and adductor magnus. The posterior fibres of gluteus medius assist with this action (Neumann 2010a). In the fully erect posture, a vertical line through the centre of gravity of

the body passes posterior to a line joining the centres of the femoral heads. The body therefore tends to incline posteriorly but this is counterbalanced by ligamentous tension and congruence and compression of the articular surfaces within the hip joints in the close-packed position. Under increased loading of the trunk or leaning posteriorly, these resistive but passive factors are assisted by active force produced by the hip joint flexors. In swaying anteriorly at the ankles, or when the arms are stretched forwards, and also in forward-bending at the hip, the line of body weight moves anterior to the medial-lateral axis of rotation through the hip joints. The posture adopted, or the rate of change of posture, is largely controlled by the hamstrings, which, besides being powerful flexors of the knee, are strong extensors of the hip. Gluteus maximus becomes particularly active when the thigh is extended against resistance, as in rising from a bending position or during climbing.

Abduction is produced by glutei medius and minimus, assisted by tensor fasciae latae, piriformis and sartorius. The motion is limited by adductor muscle tension, the pubofemoral ligament and the extreme medial bands of the descending part of the iliofemoral ligament. The abductor muscles, most notably glutei medius and minimus, are active periodically at precise phases of the walking or running cycle to ensure coronal plane stability of the pelvis.

Adduction is produced by adductors longus, brevis and magnus and by gracilis assisted by pectineus, quadratus femoris and the inferior fibres of gluteus maximus. The range of adduction is limited by the increasing tension in the abductor muscles, the transverse part of the iliofemoral ligament and the fascia lata of the thigh.

Medial rotation is produced by the anterior fibres of glutei minimus and medius, and assisted by tensor fasciae latae and most adductor muscles. The strength of medial rotation naturally increases as the hip is flexed because this position increases the moment arm of most medial rotator muscles (Delp et al 1999). Medial rotation is limited by tension in lateral rotator muscles such as piriformis, the ischiofemoral ligament and the adjacent posterior joint capsule (Wagner et al 2012).

Lateral rotation is produced by gluteus maximus, obturator internus, superior and inferior gemelli, quadratus femoris and piriformis, and it is assisted by obturator externus and sartorius. Lateral rotation, a stronger motion than medial rotation, is limited by tension in the medial rotator muscles and the transverse part of the iliofemoral ligament (Myers et al 2011).

Forces acting on the hip

In quiet upright standing, the femoral heads support the weight of the trunk, upper limbs and head. About two-thirds of body weight is located above the hips, and so each femoral head normally accepts about one-third of body weight. This force is compressive in nature, as gravity pulls the acetabula against the femoral heads. When viewed in the sagittal plane, minimal muscle forces suffice to maintain equilibrium as long as the weight of the upper body is directed over the femoral heads. If the upper body leans anteriorly, shifting the upper body weight vector anteriorly beyond the femoral heads and thereby producing a hip flexion moment, posterior thigh muscles can counter such rotation. As the capsular ligaments of the hip slacken in flexion, none of them is able to resist the forward lean.

During walking, hip joint compression force varies from one-third of body weight, when both feet are on the ground, to some four times body weight when the other foot is lifted. The large compression forces generated while walking originate primarily from two sources: gravity and muscle activation. During the stance phase, gravity pulls the upper body and acetabulum inferiorly on the femoral heads (as described above for quiet standing). More significantly, each femoral head is pushed superiorly against its socket as the foot strikes and then pushes against the ground. Although activation of muscles is responsible for a greater portion of joint compression at the hip, the influences of gravity and of capsular tension (when the hip joint is extended) should not be ignored.

As the lower limb is swinging anteriorly and free of contact with the ground, hip joint compression force is estimated to be at about 50% of body weight (Correa et al 2010). Such joint force is essentially myogenic. Consider, for example, a contracting iliopsoas pulling the femoral head against the acetabulum as it simultaneously advances the lower limb in walking.

In contrast, during the stance phase of walking, hip joint forces may reach four times body weight. These forces are mainly caused by the pull of the hip abductor muscles (primarily gluteus medius, gluteus minimus and tensor fasciae latae). This muscular action, essential to

Two terms can conveniently describe the kinematics (movements) at the hip. Femoral-on-pelvic hip kinematics describes the rotation of the femur about a relatively fixed pelvis. Conversely, pelvic-on-femoral (hip) kinematics describes the rotation of the pelvis, and often the superimposed trunk, over a relatively fixed femur or set of femurs. Regardless of whether the femur or the pelvis is the moving segment, kinematics are described for all three cardinal planes on the assumption that movements are initiated from the anatomical position.

Flexion and extension are motions that occur in the sagittal plane. To appreciate the hip joint arthrokinematics that accompany the associated femoral-on-pelvic motions, the length and angulation of the femoral neck in relation to the shaft must be considered. As the hip (femur) is flexed about 120° towards the anterior thoracic wall, or extended $10\text{--}20^\circ$ beyond the plane of the trunk, the femoral head 'spins' in the acetabulum about a side-to-side (medial-lateral) axis of rotation. Conversely, from a pelvic-on-femoral perspective, the acetabula rotate around a similar axis in flexion and extension relative to

stationary femoral heads (compare **Figs 81.12A and B**). Depending on the accompanying trunk movement, the lumbar vertebrae will either flex or extend in rhythm with the rotating pelvis. Consider, for example, a short-arc anterior rotation (or tilt) of the pelvis over both femurs while the upper trunk remains stationary. During this motion, the lumbar vertebrae extend slightly towards greater lordosis. By comparison, rotating both the pelvis and the trunk over both femurs (as in reaching towards the ground with knees held extended), the pelvis and the thoracolumbar vertebrae both flex, thereby augmenting the overall forward motion of the body as a whole.

Abduction and adduction of the hip joint occur in the coronal plane about an anterior-to-posterior axis of rotation through the femoral head. With the pelvis fixed, the femur abducts about 40° (**Fig. 81.13A**). Conversely, pelvic-on-femoral (hip) motions that occur in the coronal

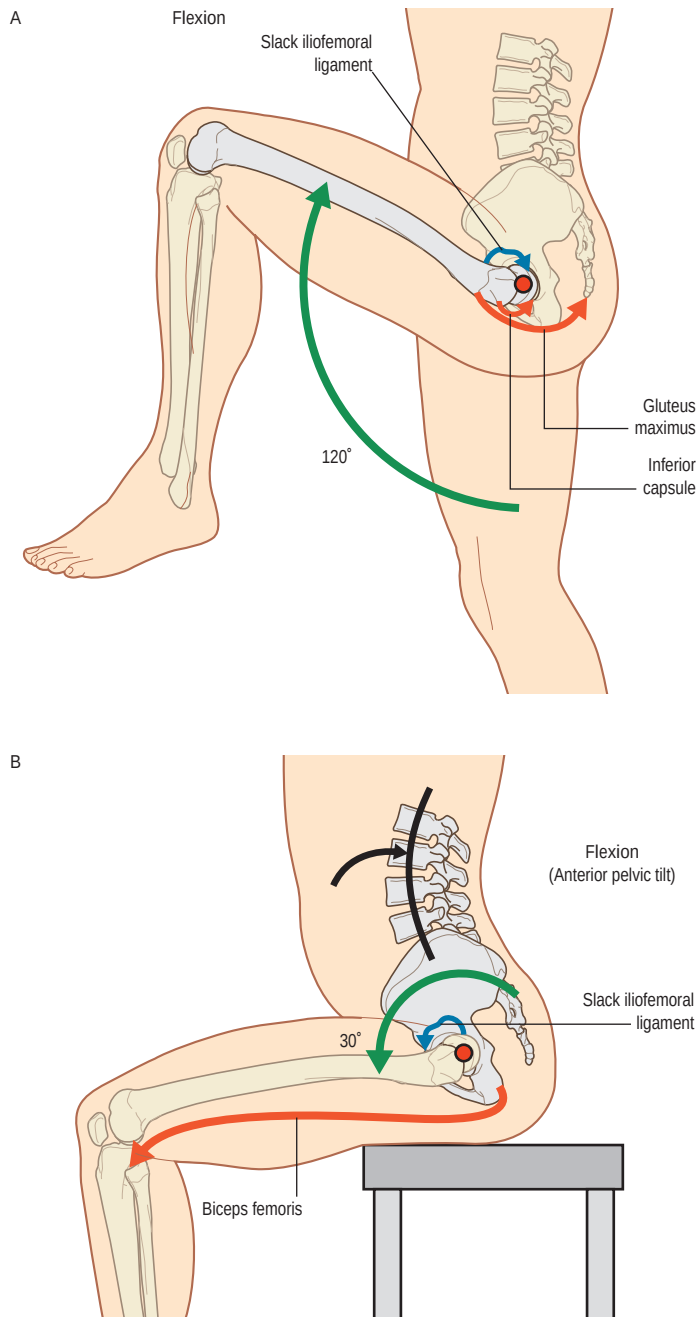


Fig. 81.12 Hip flexion shown from two kinematic perspectives, as indicated by the green curved arrows. **A**, Femoral-on-pelvic flexion. **B**, Pelvic-on-femoral flexion. Note the directional-specific motion of the lumbar spine in **B**; this is a necessity, assuming the trunk remains upright. Tissues that are elongated or pulled taut are indicated by orange arrows; tissues slackened are indicated by blue wavy arrows. In all cases, muscles are assumed to be inactive but stretched. The red circle indicates the axis of rotation at the hip. (With permission from Neumann D, *Kinesiology of the Musculoskeletal System*, 2nd ed, Elsevier, Mosby. Copyright 2010.)

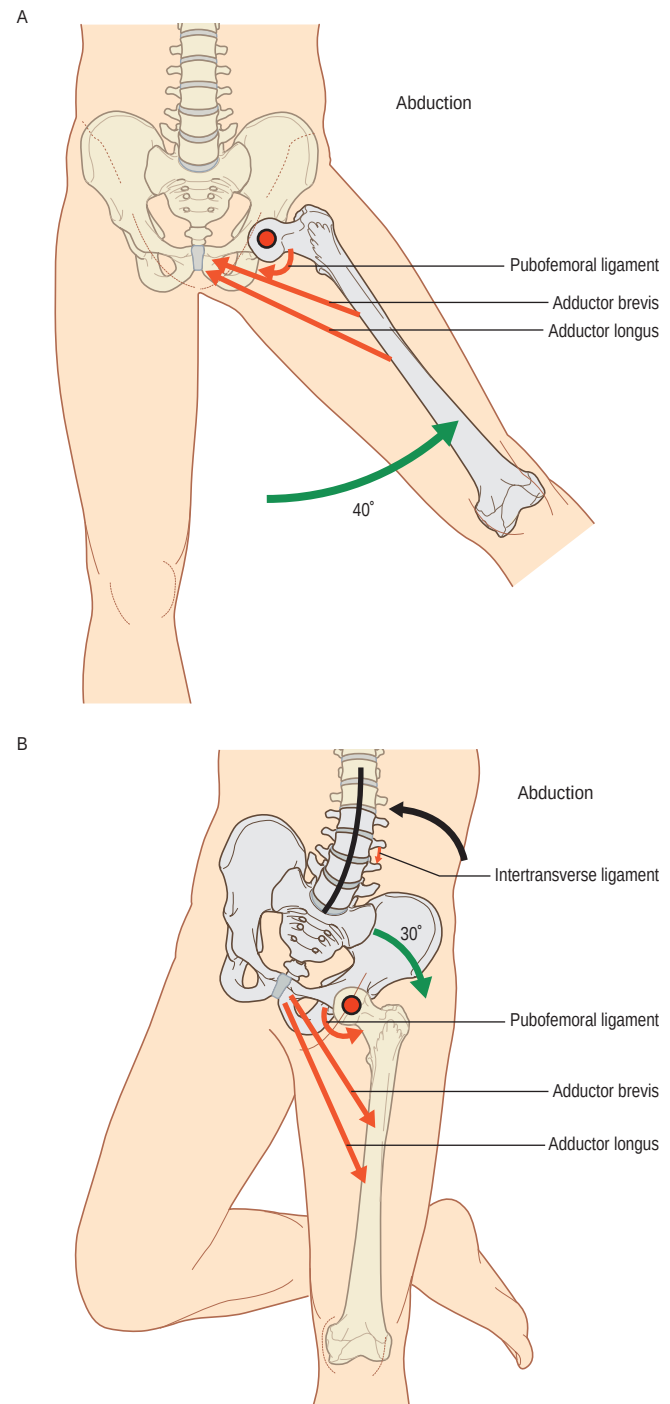


Fig. 81.13 Hip abduction shown from two kinematic perspectives, as indicated by the green curved arrows. **A**, Femoral-on-pelvic abduction. **B**, Pelvic-on-femoral abduction. Note the directional-specific motion of the lumbar spine in **B**; this is a necessity, assuming the trunk remains upright. Tissues that are elongated or pulled taut are indicated by orange arrows. In all cases, muscles are assumed to be inactive but stretched. The red circle indicates the axis of rotation at the hip. (With permission from Neumann D, *Kinesiology of the Musculoskeletal System*, 2nd ed, Elsevier, Mosby. Copyright 2010.)

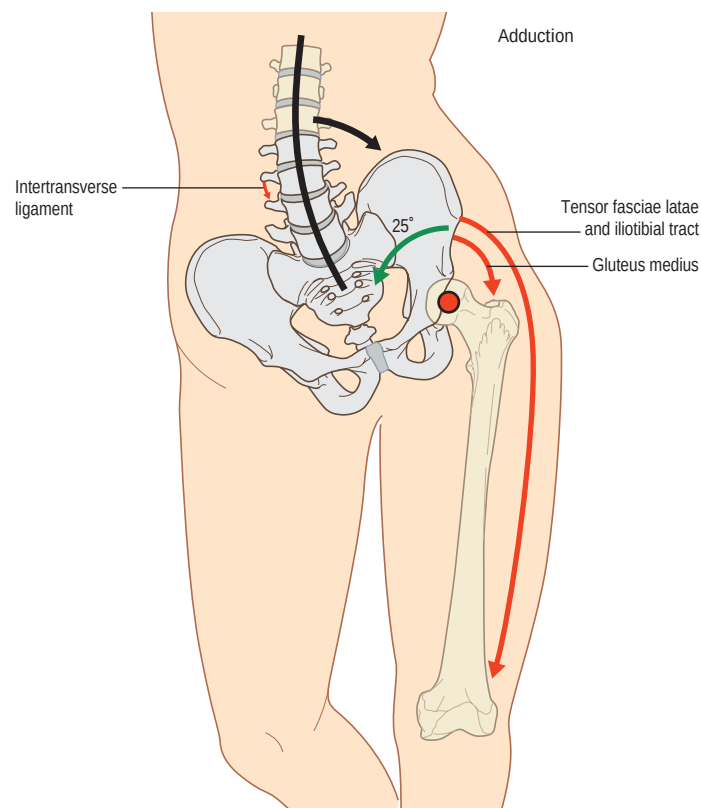


Fig. 81.14 Hip adduction shown from a pelvic-on-femoral perspective, as indicated by the green curved arrow. Note the directional-specific motion of the lumbar spine; this is a necessity, assuming the trunk remains upright. Tissues that are elongated or pulled taut are indicated by orange arrows. In all cases, muscles are assumed to be inactive but stretched. The red circle indicates the axis of rotation at the hip. (With permission from Neumann D, *Kinesiology of the Musculoskeletal System*, 2nd ed, Elsevier, Mosby. Copyright 2010.)

plane are best described assuming a person is bearing weight firmly on one lower limb, referred to hereafter as the 'support' hip. Abduction at the support hip occurs by raising the contralateral side of the pelvis a few degrees (**Fig. 81.13B**). Marked tightness in hip adductor muscles or medial capsule of the hip on the support hip can limit this pelvic motion. Adduction at the support hip occurs by lowering the contralateral side of the pelvis (**Fig. 81.14**). Assuming that the trunk remains nearly stationary during these short-arc pelvic-on-femoral motions, the lumbar spine laterally bends slightly as a way to offset undesired thoracic motion. The lumbar vertebrae, therefore, act to decouple motion of the pelvis (moving on the femur) from undesired motions between the thorax and pelvis (**Neumann 2010b**). This decoupling function of the lumbar vertebrae is especially useful during walking as a way of partially fixing the spatial orientation of the upper thorax, as well as the associated craniocervical region and the embedded senses of vision and equilibrium. This kinematic linkage between the spine, pelvis and femur exists in essentially all hip motions to varying degrees, and is an essential characteristic of normal walking.

Medial and lateral rotation of the hip joint occurs in the transverse (horizontal) plane about a vertical axis of rotation. With the pelvis

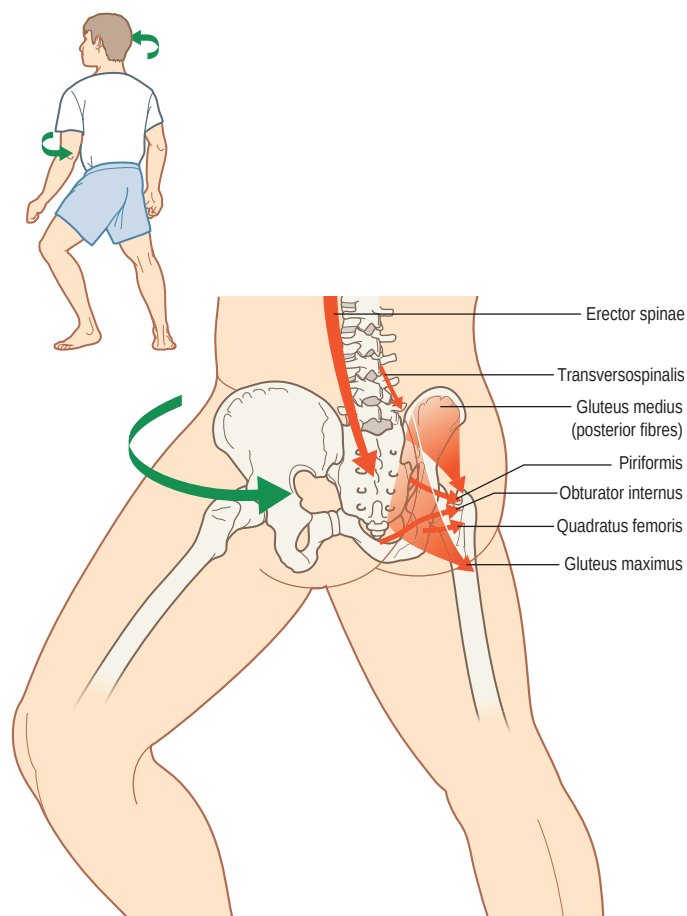


Fig. 81.15 The right lateral rotator muscles of the hip contract to produce lateral rotation of the right hip, from a pelvic-on-femoral perspective. Back extensor muscles are also shown active, rotating the lower trunk to the left. (With permission from Neumann D, *Kinesiology of the Musculoskeletal System*, 2nd ed, Elsevier, Mosby. Copyright 2010.)

fixed, the femur rotates medially about 35° . The extended hip rotates laterally about 45° . Transverse plane movements of the hip joint from a pelvic-on-femoral perspective, as explained above, are best described assuming a person is bearing weight over a given 'support' hip. Consider, for example, running and rapidly changing directions while one lower limb is securely planted on the ground. During lateral rotation of the support hip, the iliac crest located contralateral to the support limb rotates posteriorly in the transverse plane, an action driven strongly by gluteus maximus (**Fig. 81.15**). Conversely, pelvic-on-femoral (hip) medial rotation occurs as the contralateral iliac crest region rotates anteriorly.

As stated above, the axis of rotation for transverse plane motions at the hip joint occurs about a vertical axis of rotation. The slight anterior bowing for the femoral shaft, coupled with the natural angle between the femoral shaft and neck, means that this axis is not intramedullary; rather it passes just posterior and slightly medial to the shaft. The position of this 'mechanical' (versus medullary) axis becomes biomechanically relevant when considering the actions and moment arms of certain hip muscles, most notably for adductor muscles that also function as medial rotators.

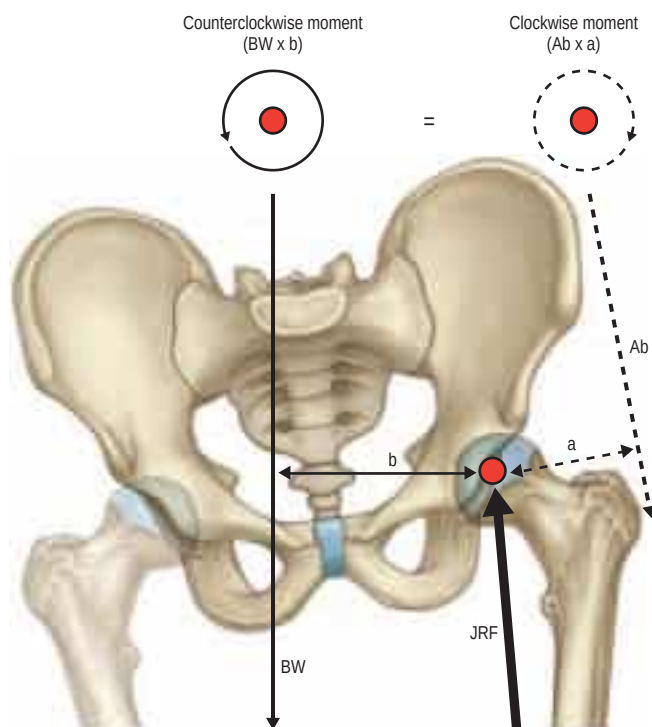


Fig. 81.16 While standing on one (left) lower limb, body weight (BW) line of force, i.e. total body weight minus the weight of the left lower limb, passes just lateral to the midline, exerting a counterclockwise moment about the stance hip ($BW \times b$). An equal but clockwise moment is required about the stance hip for mechanical equilibrium: a moment produced by the force of the abductors (Ab), whose lever arm (a) is approximately half that of the lever arm (b). The two moments are in balance when $(Ab \times a = BW \times b)$. The abductor force required to maintain equilibrium is approximately twice that of BW, resulting in a joint reaction force (JRF) about 2.5 times that of total weight of the subject's body.

normal bipedal gait, provides the necessary coronal plane stabilization of the pelvis relative to the femoral head of the stance limb.

The importance of hip abductor muscle activation during the stance phase of walking can be well appreciated by understanding the simple mechanics of standing on one limb (**Fig. 81.16**). The lever arm (a) of

left hip abductor force (Ab) is about half the length of the lever arm (b) associated with body weight (BW). To balance the competing (coronal plane) gravitational moments about the stance hip, the hip abductors must produce a force about twice superincumbent body weight. Thus, the acetabulum is pulled inferiorly against the femoral head not only by the body weight but also by the force created by the activated hip abductor muscles. The sum of the muscular and gravitational forces equals about 2.5 times the total weight of the person. These downward forces on the head of the femur are counteracted by an upward joint reaction force of equal magnitude (JRF in **Fig. 81.16**). Both magnitude and direction of the joint reaction force are strongly influenced by the pull of the hip abductor muscles.

The above analysis is for standing quietly on one limb. During walking, however, the joint reaction force routinely reaches at least four times body weight, reflecting both the high ground reaction forces and the need for the hip abductor muscles to decelerate rotations of the pelvis (**Bergmann et al 1993**). Hip joint forces reach still greater values during running or stair-climbing (see review in **Stansfield and Nicol 2002**).

Compression forces produced by the gluteus medius during stance phase are directed superiorly and slightly anteromedially within the acetabulum (**Correa et al 2010**), a region naturally protected by thick articular cartilage. In the healthy hip joint, the muscular-based compressive forces serve important functions, such as helping to stabilize the articulation and stimulate the morphological development of the growing hip. Forces that cross the hip typically do not harm the joint because they are absorbed by healthy articular cartilage and congruent joint surfaces. Failure of these conditions to absorb forces in the joint may predispose to osteoarthritis of the hip.

The three factors that influence both the magnitude and the direction of compressive forces acting on the hip are the position of the centre of gravity; the abductor moment arm, which is a function of neck length (offset) and neck-shaft angle; and the amount of body weight. Shortening of the lever arm of the abductors, such as occurs in coxa valga or excessive femoral anteversion, or in a remodelled proximal femur in advanced hip osteoarthritis, would increase abductor muscle demand and thus increase joint reaction force as well. If the muscles cannot meet this demand, the pelvis cannot be held level in standing on one limb, a problem known as either a compensated or an uncompensated Trendelenburg sign. The Trendelenburg sign is said to be compensated if the pelvis is brought towards the affected stance limb as a way of shortening the body weight's lever arm. Conversely, the Trendelenburg sign is said to be uncompensated if the pelvis drops away from the affected stance limb uncontrollably.



Bonus e-book images

Fig. 81.1B Anteroposterior radiograph of an adult male pelvis.

Fig. 81.2 T1-weighted fat-saturated MR arthrogram of the left hip joint (coronal section).

Fig. 81.3 Pre-surgical radiograph of a hip with a cam deformity of the femoral head-neck junction.

Fig. 81.8 An intact ligament of the head of the femur in a left adult hip.

Fig. 81.9 Radiograph of the left hip of a 14-year-old boy.

Fig. 81.12 Hip flexion shown from two kinematic perspectives.

Fig. 81.13 Hip abduction shown from two kinematic perspectives.

Fig. 81.14 Hip adduction shown from a pelvic-on-femoral perspective.

Fig. 81.15 The right lateral rotator muscles of the hip contract to produce lateral rotation of the right hip, from a pelvic-on-femoral perspective.

Imaging slideshow 81.1 Hip arthroscopy.

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